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Paper 1: The dawn of a new era: can machine learning and large language models reshape QS

Bibliographic Information

- **Title:** The dawn of a new era: can machine learning and large language models reshape QSP modeling?
- **Authors:** Androulakis, Ioannis P. et al.
- **Journal:** Journal of Pharmacokinetics and Pharmacodynamics, 2025, 52, 36
- **DOI:** <https://doi.org/10.1007/s10928-025-09984-5>

Paper Type: GENERIC

Quick Take

This perspective article explores how AI/ML, particularly Large Language Models (LLMs), are transitioning from simple tools to active partners in Quantitative Systems Pharmacology (QSP). It maps the impact across literature mining, dynamic model generation, surrogate/digital twin creation, and hybrid mechanistic-ML models, while outlining the immense opportunities and critical challenges, including validation and the potential of Artificial General Intelligence (AGI).

Executive Summary

This comprehensive review charts the transformative integration of artificial intelligence (AI) and machine learning (ML) within Quantitative Systems Pharmacology (QSP). The authors detail how AI/ML reshapes core QSP workflows: automating literature and data extraction, translating omics data into dynamic Boolean and ODE-based models, creating computationally efficient surrogate models for digital twins, and developing hybrid models that combine mechanistic understanding with data-driven approaches like Physics-Informed Neural Networks (PINNs). The paper's most forward-looking argument is that Large Language Models (LLMs) represent a paradigm shift, evolving from a sophisticated toolbox to an active 'partner' that can lower barriers to entry, democratize modeling, facilitate interdisciplinary communication, and even automate code generation and hypothesis testing. The discussion culminates by posing fundamental, unanswered questions about validation, regulatory oversight, ethical use, workforce evolution, and the potential future role of AGI, framing the next decade as both unpredictable and extraordinarily promising for the field.

Scientific Context & Motivation

QSP is critical for modern drug development, integrating biological knowledge and computational models to predict drug efficacy and safety. However, traditional QSP model development is laborious, requiring extensive manual literature curation, expert-driven model structuring, and computationally expensive simulations. The field faces challenges in handling the scale and

complexity of modern biological data, incorporating poorly understood biological components, and bridging the gap between quantitative modelers and life scientists. This paper addresses the emerging role of AI/ML as a solution to these bottlenecks, synthesizing recent advancements and projecting a future where AI fundamentally alters the QSP development paradigm.

✂ Methodological Snapshot

As a perspective, this paper does not present an original methodology. It presents a structured conceptual narrative. The authors' method is to synthesize and classify recent literature into four functional areas where AI/ML impacts QSP: (1) information extraction & representation, (2) data-to-dynamic-model translation, (3) surrogate model and digital twin development, and (4) hybrid mechanistic-ML model creation. They build upon this foundation to argue that modern LLMs represent a qualitative shift, enabling a fifth, more integrated paradigm of AI as a collaborative 'partner.' The analytical method is critical synthesis, culminating in a structured set of open research questions for the community.

📋 Detailed Analysis

The paper's structure effectively walks the reader through a gradient of increasing AI sophistication and integration. It begins with the straightforward application of NLP for literature mining—an efficiency gain. It then progresses to more complex tasks like network inference and Boolean-to-ODE model translation, where AI begins to define model structure. The discussion of hybrid models and PINNs represents a deep integration, where data-driven and mechanistic elements are inseparable during training. Finally, the section on LLMs argues for a culmination where the AI is an interactive agent in the scientific process, not just a processor of its data. This logical progression makes the potentially radical claim of AI as a 'partner' seem like a natural evolution. A key strength is the paper's refusal to be doctrinaire; it fully acknowledges that mechanistic components are irreplaceable for well-understood systems (e.g., basic PK), while ML excels at capturing 'dark' patterns in complex clinical data, making a compelling case for hybridism as a pragmatic, not ideological, choice. The section on digital twins is somewhat less critically developed, glossing over the monumental challenge of creating a continually 'live' model that assimilates real-time patient data—a task far exceeding the capabilities of current pharmacometric models.

Domain Context

This paper sits at the unique intersection of clinical pharmacology, systems biology, and computational science. It addresses a growing need within the pharmacometrics and QSP communities to formalize their relationship with modern AI/ML, which has often been viewed with a mixture of excitement and skepticism. It extends the conversation beyond comparing the predictive accuracy of ML vs. mechanistic models to a more holistic vision of how the entire scientific workflow—from ideation and literature review to code generation and cross-disciplinary communication—could

be transformed. This broader, process-oriented view is highly relevant for pharmaceutical R&D leadership looking to build future-ready modeling and simulation groups.

☒ Key Findings

This is a perspective piece, not a hypothesis-testing study. Its primary postulates are: 1) AI/ML can automate and enhance all stages of QSP model building, from literature mining to the direct generation of dynamic models from data. 2) Hybrid mechanistic-ML models, including PINNs and composite approaches, represent a powerful strategy that combines the interpretability of first-principles biology with the data-fitting strength of ML for poorly understood system components. 3) LLMs are shifting the paradigm from ‘tool’ to ‘partner’ by enabling researchers without deep coding expertise to build models, generate hypotheses, and translate between scientific domains through natural language. 4) The democratization of QSP by LLMs is accompanied by critical challenges in validation, reproducibility, and the risk of over-reliance and ‘hallucinations’. 5) The future may involve Artificial General Intelligence (AGI) autonomously proposing and validating models, though ethical and accountability frameworks are nascent.

Strengths & Limitations

Strengths

- Provides an exceptionally broad and well-structured synthesis of diverse AI/ML applications, connecting distinct concepts (literature mining, network inference, PINNs, digital twins, LLMs) into a single, coherent narrative about the future of QSP.
- Successfully articulates the critical transition of AI/ML from a mere ‘tool’ to a ‘partner’ via LLMs, a nuanced and provocative framing that goes beyond standard technical reviews.
- Places equal emphasis on both the transformative potential and the profound challenges (validation, hallucination, ethics, workforce), resulting in a balanced, expert-level perspective rather than uncritical advocacy.
- Concludes with a thought-provoking and detailed agenda of unanswered questions that serves as a valuable roadmap for future research, regulatory science, and professional development in the field.

Limitations (Acknowledged by Authors)

- The authors explicitly acknowledge that the work is a perspective, not a systematic review or a presentation of new experimental or computational data.
- The risk of AI/ML models generating plausible-sounding but incorrect outputs (‘hallucinations’) is discussed as a key limitation requiring mitigation.
- Challenges in ensuring rigorous model validation, regulatory compliance, and avoiding over-reliance on LLMs by non-experts are openly addressed.

Limitations (Expert Review)

- The discussion of specific AI/ML algorithms (e.g., different graph neural networks, transformer architectures beyond LLMs) remains at a high level, lacking technical depth that a computational specialist might seek.
- While challenges are listed, the paper offers no concrete, practical solutions or framework proposals for the immense validation and regulatory hurdles it identifies.
- The analysis is largely theoretical and aspirational; it does not weigh the practical success rate of, for example, direct data-to-model translation against expert-crafted models in a real-world drug development program.
- The section on digital twins does not critically address the practical gap between the concept of a continuously updated digital twin and the static, snapshot-in-time nature of most current QSP models used for regulatory decision-making.

Generalizability

As a high-level perspective, the concepts presented are broadly generalizable across all therapeutic areas where QSP is applied, including oncology, immunology, and cardiovascular disease. However, the practical applicability of each AI/ML approach (e.g., Boolean network inference from sparse clinical data vs. PINN use in data-rich preclinical settings) will be highly context-dependent.

Figures & Tables

- **Figure 1:** A conceptual diagram illustrating the four primary tasks where AI/ML can assist in QSP modeling: (1) Extracting Information, (2) Developing Surrogate Models, (3) Translating Data into Models, and (4) Developing Hybrid Mechanistic/ML Models. It also poses the central question of whether AI/ML will transition from a ‘tool’ to a ‘partner’.
 - *Significance:* This figure provides the conceptual framework for the entire paper, visually organizing the disparate applications of AI/ML into a coherent workflow and introducing the central, provocative thesis of the article.
- **Figure 2:** A schematic highlighting the challenges and opportunities at the interface of QSP, LLMs, and the user. It likely visualizes the bidirectional communication between human experts and AI partners, and the ‘magnificent challenges and opportunities’ in areas like model validation, multiscale integration, data management, and interdisciplinary collaboration.
 - *Significance:* This figure synthesizes the paper’s forward-looking discussion, emphasizing the dynamic interaction and co-evolution of the QSP scientist’s role, the LLM partner’s capabilities, and the unresolved scientific and ethical challenges.

Future Directions

The authors’ list of unanswered questions serves as its own ‘future directions’ section. Key research priorities include: establishing robust validation pipelines and benchmarking standards for LLM-assisted QSP models; developing strategies and safeguards to manage LLM ‘hallucinations’ via techniques like retrieval-augmented generation (RAG); quantifying the impact of AI-assisted

model democratization on the quality and reproducibility of scientific output; and creating ethical and regulatory frameworks for the potential future use of AGI in high-stakes pharmacological decision-making.

Expert Commentary

Androulakis et al. have produced a provocative and timely synthesis that captures a field at an inflection point. The greatest contribution is not in detailing any single method, but in elegantly reframing the conversation from ‘AI as a replacement for equations’ to ‘AI as a collaborator in the scientific process.’ The vision of LLMs bridging the language barrier between quantitative modelers and clinicians is particularly astute. However, the devil is in the details. The field’s biggest risk is a ‘gold rush’ where the power of these tools outpaces our ability to critically evaluate their outputs, a concern the authors astutely raise. This paper should serve as a call to action for regulatory scientists and methodologists to begin the hard work of building the validation and trust frameworks that will determine whether this new era fulfills its promise.

Bottom Line

For the QSP practitioner, this is a essential, high-level roadmap signaling that AI/ML literacy—from understanding hybrid model architectures to engaging with LLMs through prompt engineering—is no longer optional but is becoming foundational to the discipline. While the vision of AI as a ‘partner’ is compelling, the immediate practical message is the urgent need for the field to collectively tackle the massive challenges of validation and trust before these powerful new workflows can be responsibly deployed in regulatory drug development.

☒ Figures

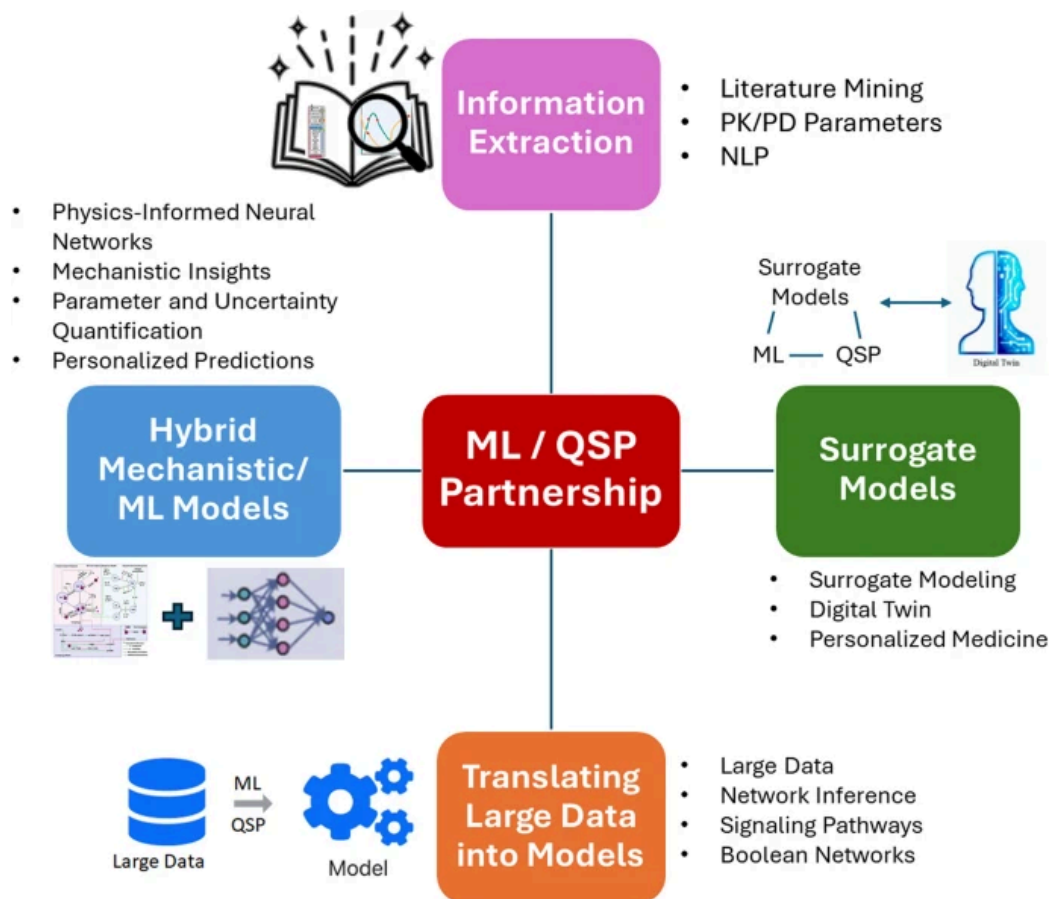


Figure 1: Figure 1

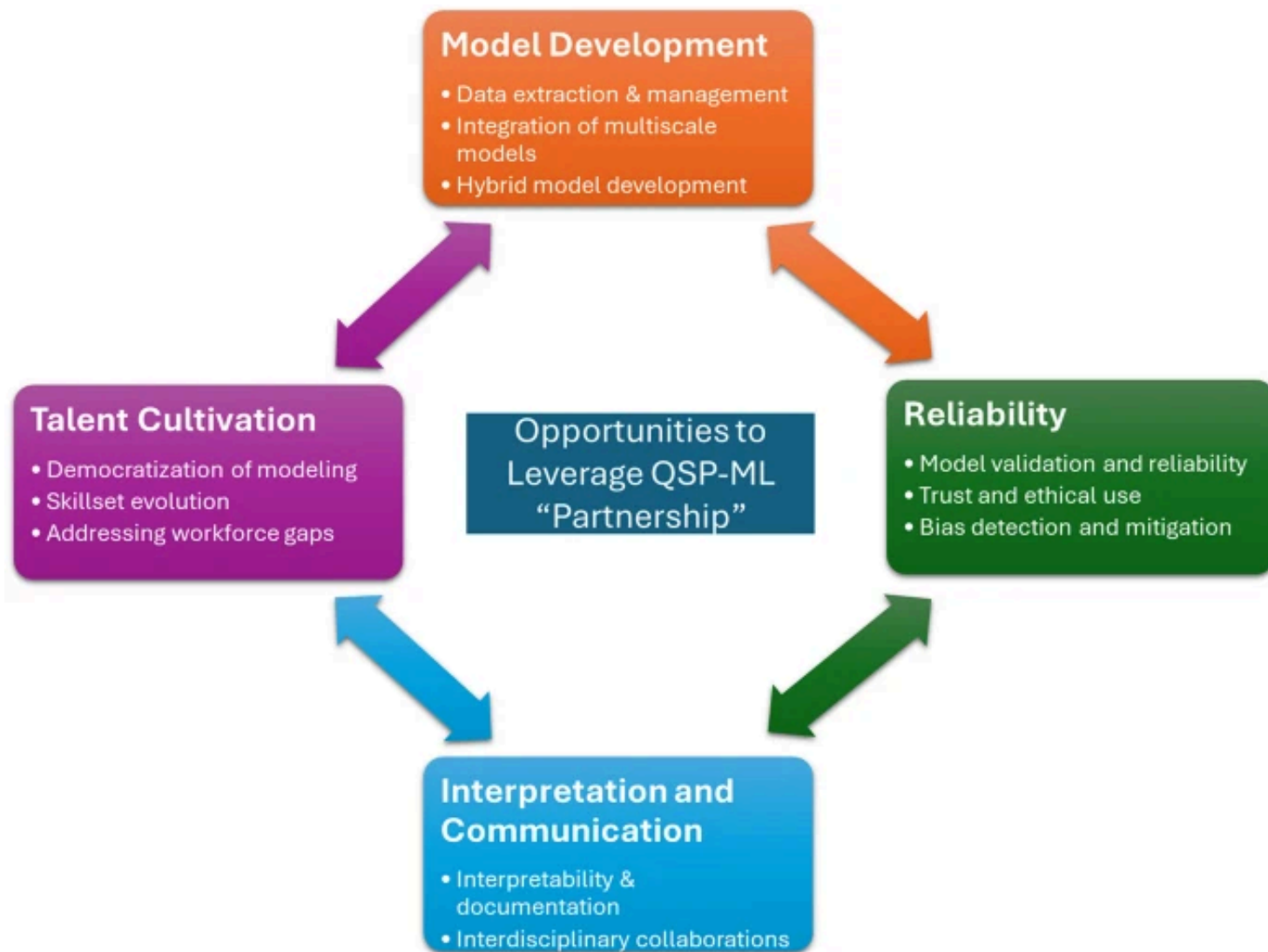


Figure 2: Figure 2